Paper-to-Peer (P2P)

How does a multi-agent system exhibit emergent behavior through synchronization?

Nov. 16. 2026 (Week 9)

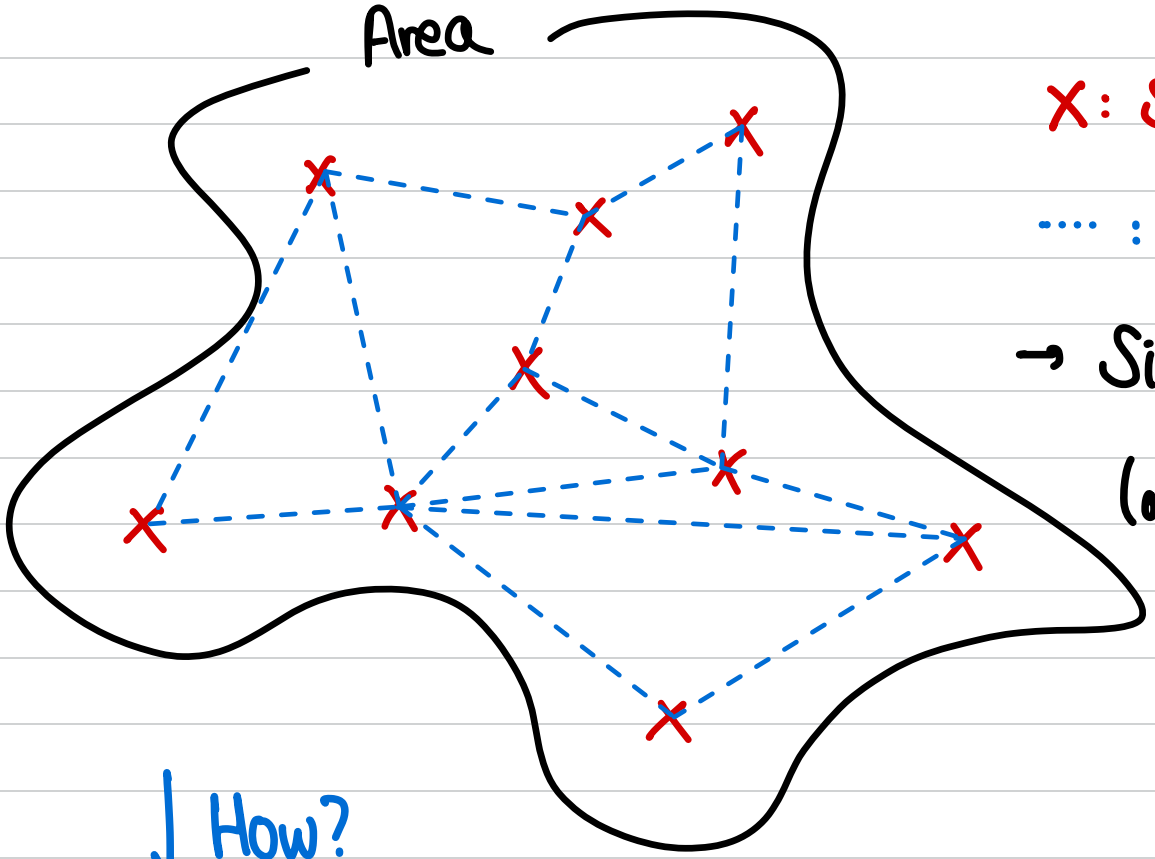
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* Review

Q. What was the major concern in the 1st presentation?

↳ Agreement (Consensus) of the network.



X : Sensors

···· : Network

→ Single Representative?

(or every node behave identically?)

↓ How?

Every node follows the "Agreement Protocol" $\dot{x}(t) = -L(G)x(t)$

Explicitly, $\dot{x}_i(t) = -\sum_{j \sim i} (x_i(t) - x_j(t))$

(using local info., every node evolves to decrease the total relative state difference.)

Thm) If every node follows the agreement protocol,

G is connected iff $x(t) \to \bar{x}_0 \mathbb{1} \in \text{span}\{\mathbb{1}\}$.

Q. How about in 2nd presentation?

"Controllability" of networked system.

↳ From any init. state, any given desired state is reachable in finite time.

Choose "input nodes" and "floating nodes"

Again, $\dot{x}(t) = -L(G)x(t)$,

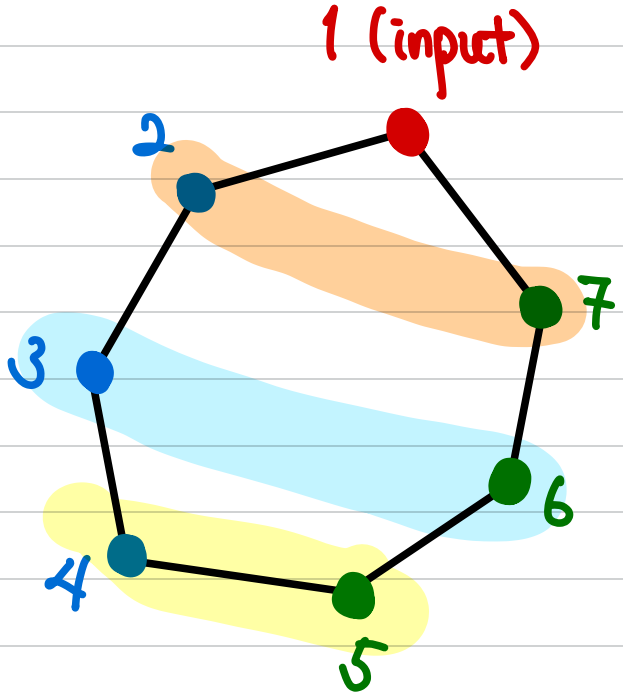
$$L(G) = D(G)D(G)^T = \begin{pmatrix} D_f \\ D_i \end{pmatrix}(D_f \; D_i) \overset{\text{let}}{=\!=} \begin{pmatrix} A_f & B_f \\ B_f^T & A_i \end{pmatrix}$$

$$\Longrightarrow \begin{pmatrix} \dot{x}_f(t) \\ \dot{x}_i(t) \end{pmatrix} = -\begin{pmatrix} A_f & B_f \\ B_f^T & A_i \end{pmatrix}\begin{pmatrix} x_f(t) \\ x_i(t) \end{pmatrix} \longrightarrow \text{LTI system}$$

$x_i(t) \equiv u(t)$ (input)

Thm) The networked system is uncontrollable if it is input symmetric.

Example : Ring graph



Command

$$x_{desired} = (1\ 2\ 3\ 4\ 5\ 6\ 7)^T$$

But, the input signal applied at 1 flows symmetrically and hence

$x(2) = x(7)$, $x(3) = x(6)$, $x(4) = x(5)$ always need to be satisfied.

↳ Command cannot be completed → Uncontrollable !

So far, each agents are identical (Same dynamics).

Q. What if the agents are Heterogeneous ?

How do they behave ?

↓ move our concern back to consensus.

$$\text{Consensus} = \text{Synchronization} = \lim_{t \to \infty} x(t) \in \text{span}\{\mathbb{1}\}.$$

Q1. How consensus give rise to emergent behavior?

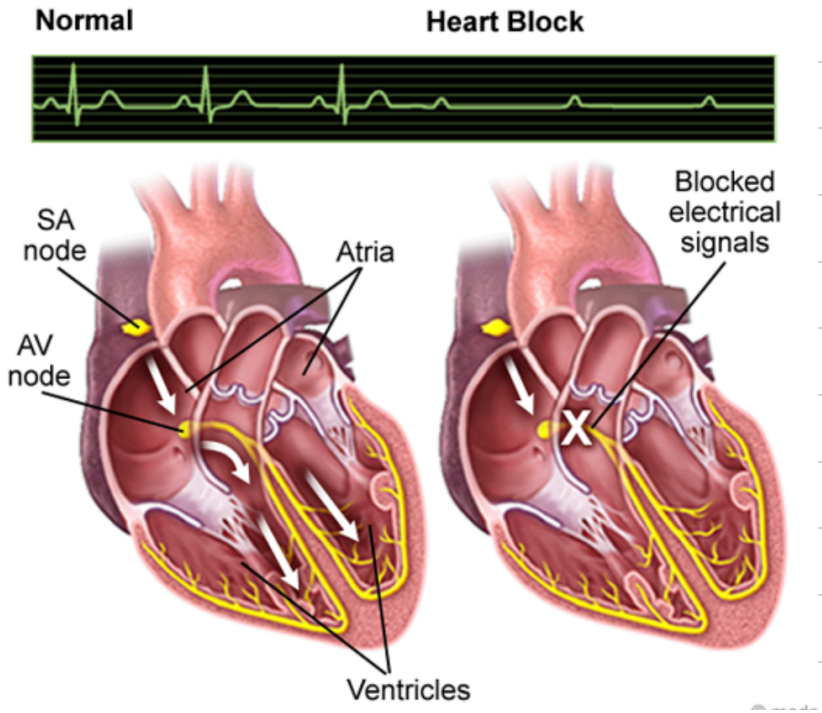
Q2 How consensus give rise to robustness of group behavior?

Consensus = Synchronization = $\lim_{t\to\infty} x(t) \in \text{span}\{\mathbb{1}\}$.

Q1. How consensus give rise to emergent behavior?

Q2 How consensus give rise to robustness of group behavior?

- Biological organ (in nature)



thousands of pacemaker cells
having different periods of firing

↓ synchro

One common Heartbeat ratio

$$\dot{x}_i = f_i(t, x_i) + k\sum_{j\in N_i} I_{ij} \; ; \; I_{ij} = g_{ij}(V_i - V_j)$$

Consensus = Synchronization = $\lim_{t\to\infty} x(t) \in \text{span}\{\mathbb{1}\}$.

Q1. How consensus give rise to emergent behavior?

Q2 How consensus give rise to robustness of group behavior?

- Networked Robot (intentional design)



Drones with different dynamics

$x_i(t)$ · position, $y_i := x_i - d_i$.

↓ synchro.

$y_i \to y_j \iff x_i - x_j \to d_i - d_j$

(formation)

* Preliminary : Singular perturbation theory.

Motivating example :

$$\begin{cases} \dot{x}(t) = -z(t) & ; \ x(0) = -1 \\ \dot{z}(t) = -1000\,(z(t) - x(t)) & ; \ z(0) = 1. \end{cases}$$

Q. How $x(t)$ and $z(t)$ behaves ?

- Observation : $\dot{x}(0) = 1$, $\dot{z}(0) = -2000$

  → z will show drastic behavior near $t = 0$.

- Observation 2 : $\dot{z}(t) = -1000\,(z(t) - x(t))$

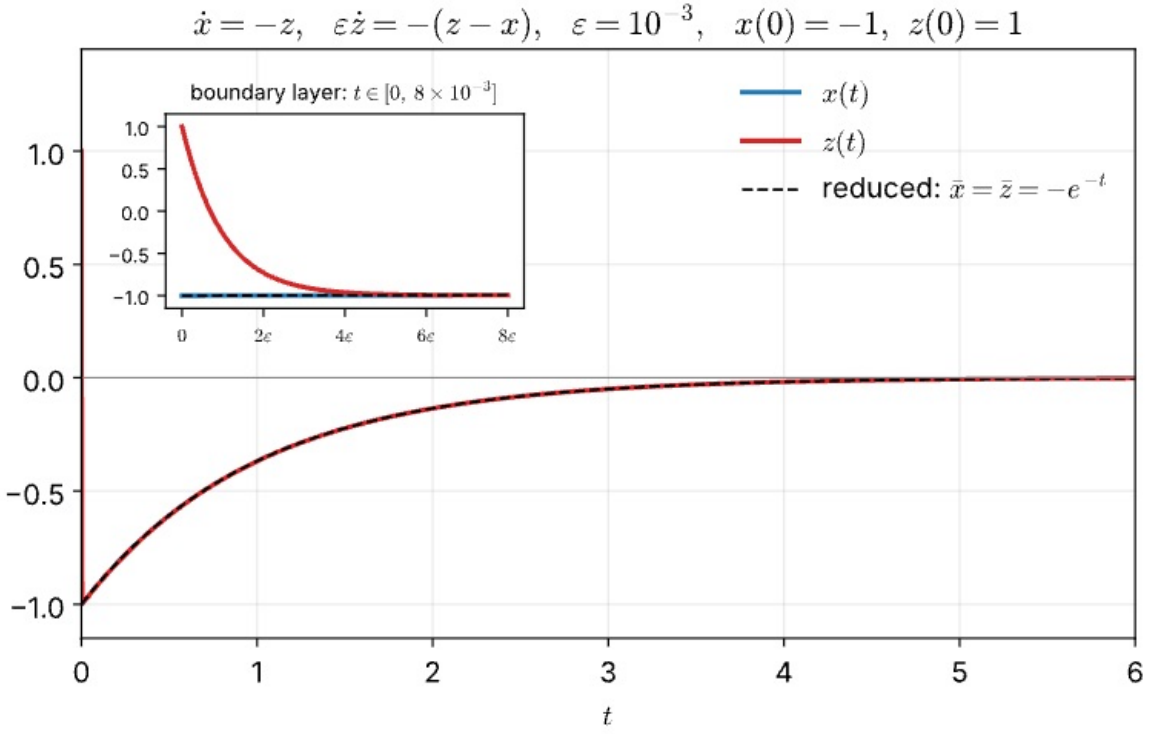
  → z moves "toward" x

"$z(t)$ rapidly approaches to $x(t)$!"

Extremely, assume that $x(t) = z(t)$ for all $t > 0$

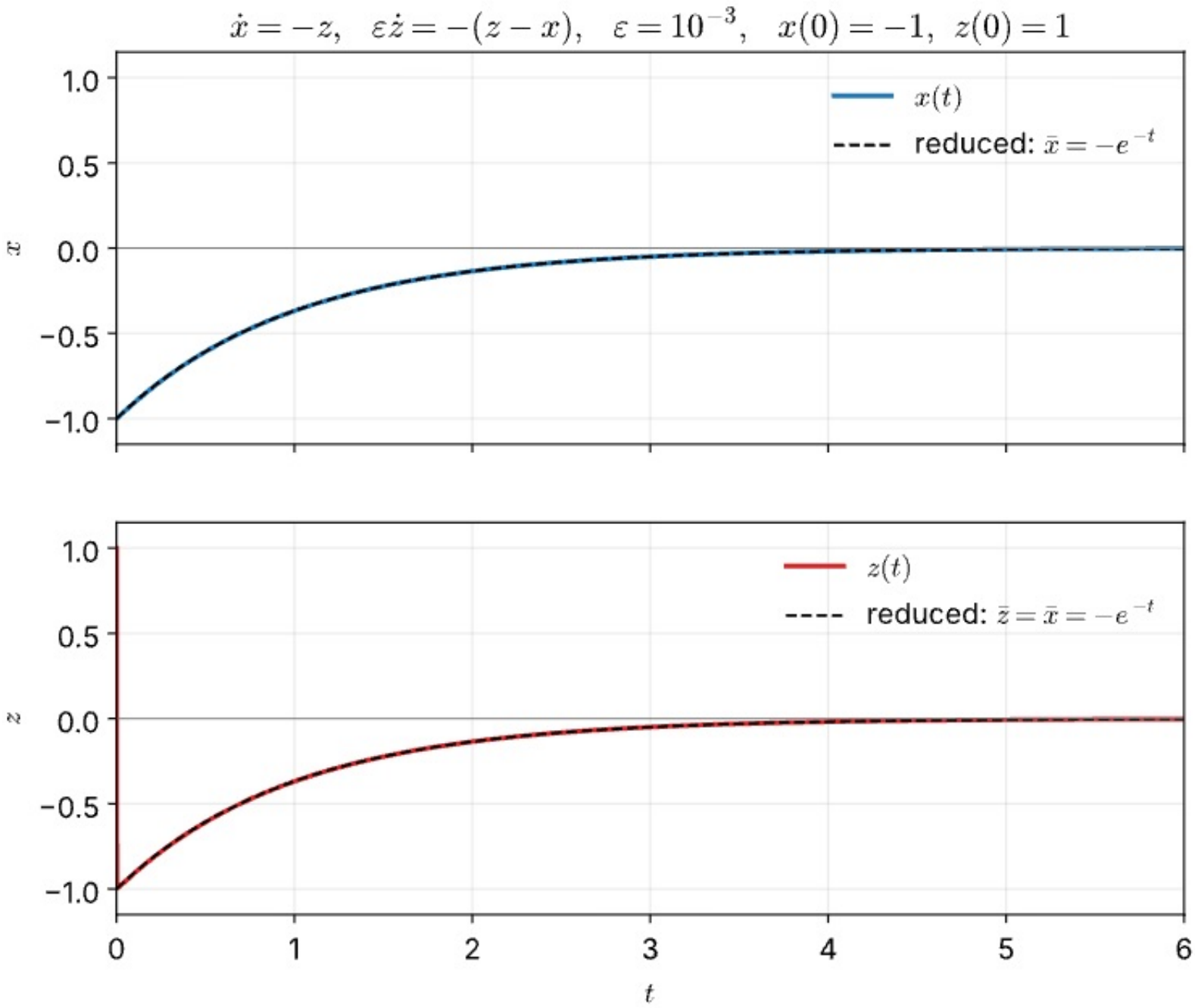
$$\rightarrow \begin{cases} \dot{x}(t) = -x(t), \quad x(0) = -1 \\ z(t) = x(t) \end{cases} \Rightarrow \begin{array}{l} x(t) = -e^{-t} \\ z(t) = -e^{-t} \end{array}$$

Q. Is this extreme assumption plausible?



As we expected,

- $z(t) \rightarrow x(t)$ rapidly
- $z(t) \approx x(t) \quad \forall t > 0$.


$\dot{x} = -z, \quad \varepsilon\dot{z} = -(z - x), \quad \varepsilon = 10^{-3}, \quad x(0) = -1, \; z(0) = 1$
$x(t)$
reduced: $\bar{x} = -e^{-t}$
x
$z(t)$
reduced: $\bar{z} = \bar{x} = -e^{-t}$
z
t

Then ..

System = Fast dynamics + Slow dynamics



we can analyze the "reduced" system

rather than analyzing the entire system.

* Standard singular perturbation form

$$\begin{cases} \dot{x} = f(t, x, z, \varepsilon) & \text{(Slow part)} \\ \varepsilon \dot{z} = g(t, x, z, \varepsilon) & \text{(fast part)} \end{cases}$$

Assume that f, g continuously differentiable for

$(t, x, z, \varepsilon) \in [0, t_1] \times D_x \times D_z \times [0, \varepsilon_0]$ where $D_x \subset \mathbb{R}^n$, $D_z \subset \mathbb{R}^m$

① Let $\varepsilon = 0$ and solve $0 = g(t, x, z, 0)$. $\Rightarrow$ Obtain $z = h(t, x)$

② Substitute this to the slow part $\Rightarrow$ $\dot{x} = f(t, x, h(t, x), 0)$

Original system : $n + m$ order

Reduced system : n order

* Analysis of Heterogeneous multi-agent system.

Heterogeneous dynamics : $\dot{x}_i(t) = f_i(t, x_i) + u_i$, $\forall i \in [N]$

- $x_i(t) \in \mathbb{R}^n$: State of agent i
- $f_i : [0, \infty) \times \mathbb{R}^n \to \mathbb{R}^n$: vector field w.r.t. agent i.
- u_i : Control input to state x_i → interactions with other agents.

Assume Static diffusive state coupling : $u_i = k \sum_{j=1}^{N} a_{ij}(x_j - x_i)$

- k: Coupling gain
- $[a_{ij}]$: Adjacency matrix

So, in short, $u = -kLx$ where $x = (x_1^T \cdots x_n^T)^T$, $u = (u_1^T \cdots u_n^T)^T$

Question 1.

How consensus give rise to emergent behavior?

Q1. How consensus give rise to emergent behavior?

- Change dynamics into singular perturbation form.

Linear coordinate change: $\Phi : \mathbb{R}^{nN} \to \mathbb{R}^{nN} : x \mapsto \begin{pmatrix} s \\ z \end{pmatrix} = \left(\begin{pmatrix} \frac{1}{N} \mathbf{1}_N^T \\ R^T \end{pmatrix} \otimes I_n \right) x$

$$s = \frac{1}{N}(\mathbf{1}_N \otimes I_n)x = \frac{1}{N}\sum_{i=1}^{N} x_i \in \mathbb{R}^n, \qquad z = (R^T \otimes I_n)x \in \mathbb{R}^{(N-1)n}$$

where $R \in \mathbb{R}^{N \times (N-1)}$ is any matrix satisfying

$$\begin{pmatrix} \frac{1}{N} \mathbf{1}_N^T \\ R^T \end{pmatrix} \mathcal{L}(G) \begin{pmatrix} \mathbf{1}_N & R \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \Lambda \end{pmatrix}$$

with $\Lambda \succeq 0$, $\Lambda \in \mathbb{R}^{(N-1)\times(N-1)}$.

By letting $x = (x_1^T, \cdots, x_N^T)^T \in \mathbb{R}^{nN}$

$F(t,x) = (f_1(t,x_1)^T, \cdots, f_N(t,x_N)^T)^T \in \mathbb{R}^{nN}$

$\Big\rangle \; \dot{x} = F - k(L \otimes I_n)x$

and we have $s = \frac{1}{N}(\mathbb{1}_N^T \otimes I_n)x$, $\tilde{z} = (R^T \otimes I_n)x$.

$\Rightarrow \; x = (\mathbb{1}_N \otimes I_n)s + (R \otimes I_n)\tilde{z}$, $x_i = s + (R_i \otimes I_n)\tilde{z}$

$$\dot{s} = \frac{1}{N}(\mathbb{1}_N^T \otimes I_n)\dot{x} = \frac{1}{N}(\mathbb{1}_N^T \otimes I_n)(F - k(L \otimes I_n)x)$$

$$= \frac{1}{N}(\mathbb{1}_N^T \otimes I_n)F - \frac{k}{N}\underbrace{(\mathbb{1}_N^T \otimes I_n)(L \otimes I_n)}_{\| \; \because (A \otimes B)(C \otimes D) = (AC) \otimes (BD)}x$$

$$(\mathbb{1}_N^T L) \otimes I_n$$

$$\| \; 0$$

$$\dot{\tilde{z}} = (R^T \otimes I_n)\dot{x} = (R^T \otimes I_n)(F - k(\mathbb{1}\otimes I_n)x)$$

$$= (R^T \otimes I_n)F - k(R^T \otimes I_n)(\mathbb{1}\otimes I_n)x$$

$$= (R^T \otimes I_n)F - k(R^T\mathbb{1}\otimes I_n)((\mathbb{1}_N \otimes I_n)S + (R\otimes I_n)\tilde{z})$$

$$\downarrow$$

$$\cancel{(R^T\mathbb{1}\otimes I_n)(\mathbb{1}_N\otimes I_n)S} + (R^T\mathbb{1}\otimes I_n)(R\otimes I_n)\tilde{z}$$

$\because (A\otimes B)(C\otimes D) = (AC)\otimes(BD)$

$$\| \qquad\qquad \|$$

$$(R^T\mathbb{1}\mathbb{1}_N^T)\otimes I_n \qquad (R^T\mathbb{1}R)\otimes I_n$$

$$\| \qquad\qquad \|$$

$$0 \qquad\qquad \Lambda$$

$$= (R^T \otimes I_n)F - k(\Lambda\otimes I_n)\tilde{z}$$

Standard Singular perturbation form of dynamics $\dot{x} = F - k(L \otimes I_n)x$

$$\begin{cases} \dot{s} = \frac{1}{N}(\mathbb{1}_N^T \otimes I_n)F = \frac{1}{N}\sum_{i=1}^{N} f_i(t, s + (R \otimes I_n)\tilde{z}) & \text{(Slow part)} \\ \frac{1}{k}\dot{\tilde{z}} = -(\Lambda \otimes I_n)\tilde{z} + \frac{1}{k}(R^T \otimes I_n)F & \text{(fast part)} \end{cases}$$

By letting $\tilde{z} = 0$, we obtain $\boxed{\dot{s} = \frac{1}{N}\sum_{i=1}^{N} f_i(t, s) \quad \text{(Blended dynamics)}}$

Recall that $x_i = s + (R_i \otimes I_n)\tilde{z}$ and we expect that the behavior of multi-agent system can be approximated by the blended dynamics

Thm) Blended dynamics theorem (general statement)

When heterogeneous agents are strongly coupled through diffusive coupling, each agent's behavior is approximated by the blended dynamics, the average of the individual vector fields.

If the blended dynamics is stable, then this approximation holds for all time, and the error can be made arbitrarily small by increasing the coupling gain

From the above statement, we emphasize

① Stability is required only for blended dynamics, not for individuals.

② Blended dynamics shows an emergent behavior of MAS

* Assumptions

In this presentation, we assume $n=1$ (scalar) for simplicity and impose the following 3 conditions.

① Each vector field $f_i(t, x_i)$ is

- piecewise continuous in t, C^1 in x_i ;
- $f_i(t, 0)$ uniformly bounded for t ;
- locally Lipschitz w.r.t. x_i uniformly in t.

$\exists$ nondecreasing continuous ftn. $M: \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ and a constant $L > 0$ s.t. $\forall x_i \in \mathbb{R}$, $\forall t \geq 0$, $|f_i(t, x_i)| \leq M(|x_i|)$ and $\left|\frac{\partial f_i}{\partial x_i}(t, x_i)\right| \leq L$

② Underlying network is given by a Connected, undirected graph.

③ Stability of blended dynamics

There is a constant $p > 0$ s.t. for all $s \in \mathbb{R}$, $t \geq 0$,

$$\frac{\partial \bar{f}}{\partial s}(t, s) = \frac{1}{N} \sum_{i=1}^{N} \frac{\partial f_i}{\partial x_i}(t, s) \leq -p$$

Theorem 1) Blended dynamics theorem (Kim, 2016)

Under assumptions 1, 2, and 3, there exists a class $\mathcal{K}$-function γ_N

s.t. the solutions of the overall system $\dot{x} = f(t,x) - kLx$

with arbitrary initial conditions, and the solution of blended dynamics

$\dot{s} = \frac{1}{N}\sum_{i=1}^{N} f_i(t,s)$ with initial condition $s(0) = \bar{x}(0)$ satisfy

$$\limsup_{t\to\infty} |x_i(t) - s(t)| \leq \gamma_N\left(\frac{1}{k\lambda_2 - L}\right), \quad \forall k \geq \bar{K}, \ \forall i \in [N].$$

where $\bar{K} = \frac{3L^2}{P\lambda_2} + \frac{L}{\lambda_2}$.

Lemma 1

For a scalar system $\dot{s} = F(t,s)$ where F is C^1 function s.t.

$\frac{\partial F}{\partial s} \leq -p < 0$ for all s and nonnegative t,

$$\limsup_{t\to\infty} |s(t)| \leq \frac{1}{p} \limsup_{t\to\infty} |F(t,0)|$$

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pf) By MVT, $F(t,s) - F(t,0) = \frac{\partial F}{\partial s}(t,\xi)s$ for some $\xi \in [0,s]$

And since $\frac{d|s|}{dt} = \operatorname{sgn}(s)\,\dot{s} = \operatorname{sgn}(s)F(t,s)$,

$$\frac{d|s|}{dt} = \operatorname{sgn}(s)\left(F(t,0) + \underbrace{\frac{\partial F}{\partial s}(t,\xi)\cdot s}_{\leq -p}\right) \leq |F(t,0)| - p|s(t)| \quad (*)$$

Next, consider a linear ODE $\dot{\omega} = -p\omega + |F(t,0)|$, $\omega(0) = |S(0)|$

By comparison lemma, together with (*), $|S(t)| \le \omega(t)$ $\forall t$.

Solve

Here, since $\omega(t) = |S(0)|e^{-pt} + \int_0^t e^{-p(t-\tau)}|F(\tau,0)|\,d\tau$

Let $c = \limsup_{t\to\infty} |F(t,0)|$, i.e. for any $\delta > 0$, $\exists T$ s.t.

$$|F(\tau,0)| \le c+\delta \quad \forall \tau > T$$

$\int_0^t = \int_0^T + \int_T^t$

Then, $\omega(t) = |S(0)|e^{-pt} + \int_0^T e^{-p(t-\tau)}|F(\tau,0)|\,d\tau + \int_T^t e^{-p(t-\tau)}|F(\tau,0)|\,d\tau$

$|S(0)|e^{-pt} \to 0$ as $t\to\infty$; $\int_0^T e^{-p(t-\tau)}|F(\tau,0)|\,d\tau \to 0$ as $t\to\infty$; $|F(\tau,0)| \le c+\delta$

and hence, $\omega(t) \le \int_T^t e^{-p(t-\tau)}(c+\delta)\,d\tau = \frac{c+\delta}{p}(1-e^{-p(t-\tau)}) \le \frac{c+\delta}{p}$.

Therefore, $\limsup_{t\to\infty} \omega(t) \le \frac{c+\delta}{p}$ for any δ. by letting $\delta \to 0$, proved ■

Lemma 2

Let $\rho_k(x,y) = -\begin{pmatrix} |x| & |y| \end{pmatrix}^T \begin{pmatrix} p & a \\ a & k \end{pmatrix} \begin{pmatrix} |x| \\ |y| \end{pmatrix} + \theta(t)|y|$

where $x \in \mathbb{R}^l$, $y \in \mathbb{R}^m$, $\theta(t) \geq 0$ and $a, p > 0$ are constants.

Then, there exists a class $\mathcal{K}$-function r and a constant $c > 0$ s.t.

$$\rho_k(x,y) \leq -c(|x|^2 + |y|^2) \quad \text{if } |x|^2 + |y|^2 > \theta(t) r(1/k) \quad \forall k > 3a^2/p.$$

Proof of theorem 1) Step 1: Setting

Recall that we introduced a linear transform $\Phi(x) = \begin{pmatrix} \frac{1}{N}\mathbf{1}_N^T \\ R^T \end{pmatrix} x$ where

$R \in \mathbb{R}^{N \times (N-1)}$ is any matrix satisfying $\begin{pmatrix} \frac{1}{N}\mathbf{1}_N^T \\ R^T \end{pmatrix} \mathcal{L}(G) \begin{pmatrix} \mathbf{1}_N & R \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \Lambda \end{pmatrix}$

with $\Lambda \succeq 0$, $\Lambda \in \mathbb{R}^{(N-1)\times(N-1)}$.

Let $\{\lambda_1, \cdots, \lambda_N\}$ and $\{u_1, \cdots, u_N\}$ be the set of eigenvalues and orthonormal eigenvectors of $\mathcal{L}(G)$.

By choosing $R = (u_2\ u_3 \cdots u_N)$, then $\Lambda = R^T \mathcal{L}(G) R = \mathrm{diag}(\lambda_2, \lambda_3, \cdots, \lambda_N)$

Here, by the connectivity of G, $\lambda_1 = 0 < \lambda_2 \leq \cdots \leq \lambda_N$, Λ is PSD.

Then, we have some properties

$$R^T R = I_{N-1} \quad \text{(by orthonormality)}$$

$$R^T \mathbb{1}_N = \begin{pmatrix} u_1^T \\ \vdots \\ u_N^T \end{pmatrix} \mathbb{1}_N = \begin{pmatrix} u_2^T \mathbb{1}_N \\ \cdot \\ u_N^T \mathbb{1}_N \end{pmatrix} = 0 \quad (\because u_1 = \mathbb{1}, \text{ and orthonormality})$$

$$\|R\| = 1$$

* Step 2: Error dynamics

Let $\Phi(x) = Wx = \begin{pmatrix} \xi_1 \\ \tilde{\xi} \end{pmatrix}$. Then, we have a singular perturbation form

$$\begin{cases} \dot{\xi}_1 = \frac{1}{N} 1_N^T f(t, 1_N \xi_1 + R\tilde{\xi}) & \text{(Slow term)} \\ \frac{1}{k}\dot{\tilde{\xi}} = -\Lambda \tilde{\xi} + \frac{1}{k} R^T f(t, 1\xi_1 + R\tilde{\xi}) & \text{(fast term)} \end{cases}$$

Now, define $e = \xi_1 - s$ where $s(t)$ is a blended dynamics.

By subtracting $\dot{s} = \frac{1}{N} 1_N^T f(t, 1s)$ to the slow term,

$$\dot{e} = \dot{\xi}_1 - \dot{s} = \frac{1}{N} 1_N^T f(t, 1_N \xi_1 + R\tilde{\xi}) - \frac{1}{N} 1_N^T f(t, 1s) \quad \text{(Error dynamics)}$$

* Step3-1 : Lyapunov analysis - Separating terms

Define $V = \frac{1}{2}(Ne^2 + |\tilde{\xi}|^2)$, and $v = \sqrt{N}|e|$ so that $2V = v^2 + |\tilde{\xi}|^2$.

Consider $|x - \mathbb{1}s|$ Since $\mathbb{1}\xi_1 + R\tilde{\xi} = x$ and $\xi_1 = e + s$,

$$|x - \mathbb{1}s|^2 = |\mathbb{1}e + R\tilde{\xi}|^2 = (\mathbb{1}e + R\tilde{\xi})^T(\mathbb{1}e + R\tilde{\xi})$$

$$= e^T\mathbb{1}^T\mathbb{1}e + \underbrace{2e\mathbb{1}^TR\tilde{\xi}}_{=0} + \tilde{\xi}^T\underbrace{R^TR}_{=I}\tilde{\xi}$$

$$= Ne^2 + |\tilde{\xi}|^2$$

$$= 2V$$

$$\rightarrow |x - \mathbb{1}s| = \sqrt{2V}.$$

Note that $\dot{V} = Ne\dot{e} + \tilde{\xi}^T\dot{\tilde{\xi}}$

For the first term, by substituting error dynamics,

$$Ne\dot{e} = Ne\left(\frac{1}{N}\mathbb{1}_N^T f(t, \mathbb{1}_N\xi_1 + R\tilde{\xi}) - \frac{1}{N}\mathbb{1}_N^T f(t, \mathbb{1}_N s)\right)$$

$$= e\mathbb{1}^T f(t, \mathbb{1}\xi_1 + R\tilde{\xi}) - \mathbb{1}^T f(t, \mathbb{1}\xi_1) + \mathbb{1}^T f(t, \mathbb{1}\xi_1) - e\mathbb{1}^T f(t, \mathbb{1}s)$$

$$= e\mathbb{1}^T\left(f(t, \mathbb{1}\xi_1 + R\tilde{\xi}) - f(t, \mathbb{1}\xi_1)\right) - e\mathbb{1}^T\left(f(t, \mathbb{1}\xi_1) - f(t, \mathbb{1}s)\right)$$

Similarly, for the second term,

$$\tilde{\xi}^T\dot{\tilde{\xi}} = \tilde{\xi}^T\left(-k\Lambda\tilde{\xi} + R^T f(t, \mathbb{1}\xi_1 + R\tilde{\xi})\right)$$

$$= -k\tilde{\xi}^T\Lambda\tilde{\xi} + \xi^T R^T f(t, \mathbb{1}\xi_1 + R\tilde{\xi}) + \xi^T R^T f(t, \mathbb{1}s) - \xi^T R^T f(t, \mathbb{1}s)$$

$$= -k\tilde{\xi}^T\Lambda\tilde{\xi} + \xi^T R^T f(t, \mathbb{1}s) + \tilde{\xi}^T R^T\left(f(t, \mathbb{1}\xi_1 + R\tilde{\xi}) - f(t, \mathbb{1}s)\right)$$

* Step 3-2. Term-by-term bound

Note: Component-wise MVT.

$f(t,Y) - f(t,Z) = D(Y-Z)$ where $D = \mathrm{diag}\left(\frac{\partial f_1}{\partial x_1}(t,\omega_1), \cdots, \frac{\partial f_N}{\partial x_N}(t,\omega_N)\right)$

for some $Y_i \le \omega_i \le Z_i \quad \forall i \in [N]$

(1) $e\mathbf{1}^T\left(f(t, \mathbf{1}\xi_1 + R\tilde{\xi}) - f(t, \mathbf{1}\xi_1)\right) \overset{\text{let}}{=} \alpha$

By applying componentwise MVT, $f(t, \mathbf{1}\xi_1 + R\tilde{\xi}) - f(t, \mathbf{1}\xi_1) = D(R\tilde{\xi})$

and by assumption that each f_i is L-Lipschitz,

$$\|\alpha\| \le \|e\| \cdot \underbrace{\|\mathbf{1}\|}_{\sqrt{N}} \cdot \underbrace{\|D\|}_{L} \cdot \underbrace{\|R\|}_{1} \|\tilde{\xi}\| \le |e|\sqrt{N} \cdot L|\tilde{\xi}| = L\nu|\tilde{\xi}|$$

(2) $e\mathbb{1}^T(f(t,\mathbb{1}s_1) - f(t,\mathbb{1}s)) \overset{\text{let}}{=} \beta$

Since $\bar{f}(t,s) = \frac{1}{N}\mathbb{1}^T f(t,\mathbb{1}s)$, $\beta = Ne(\bar{f}(t,s+e) - \bar{f}(t,s))$

By applying MVT, for some $g \in (s, s+e)$,

$$\beta = Ne\frac{\partial \bar{f}}{\partial s}(t,g)\cdot e = Ne^2\frac{\partial \bar{f}}{\partial s}(t,g)$$

By the third assumption, $\frac{\partial \bar{f}}{\partial s} \leq -p$ and hence $\beta \leq -pNe^2 = -pv^2$.

(3) $-k\tilde{\xi}^T\Lambda\tilde{\xi} \overset{\text{let}}{=} \gamma$

Since $\lambda_2 \leq \lambda_3 \leq \cdots \leq \lambda_N$, $\Lambda - \lambda_2 I \succeq 0$.

So, $\gamma \leq -k\lambda_2|\tilde{\xi}|^2$

(4) $\tilde{\xi}^T R^T f(t, \mathbb{1}s) \overset{\text{let}}{=} \delta$

By letting $\theta(t) := |R^T f(t, \mathbb{1}s)|$ and applying Cauchy-Schwartz ineq.,

$$\delta \leq \theta(t)|\tilde{\xi}|.$$

(5) $\tilde{\xi}^T R^T (f(t, \mathbb{1}\xi_1 + R\tilde{\xi}) - f(t, \mathbb{1}s)) \overset{\text{let}}{=} \eta$

Again, by MVT, $f(t, \mathbb{1}\xi_1 + R\tilde{\xi}) - f(t, \mathbb{1}s) = D(\mathbb{1}e + R\tilde{\xi})$.

Then, $\|\eta\| \leq \|\tilde{\xi}\| \, \|R\| \cdot \|D\| \cdot \|\mathbb{1}e + R\tilde{\xi}\|$

$$\leq |\tilde{\xi}| L \cdot (\nu + |\tilde{\xi}|) = L\nu|\tilde{\xi}| + L|\tilde{\xi}|^2.$$

Thus, $\dot{V} \leq -p\nu^2 + 2L\nu|\tilde{\xi}| - (k\lambda_2 - L)|\tilde{\xi}|^2 + \theta(t)|\tilde{\xi}|.$

* Step 3-3 : Applying lemma 2.

By letting $\kappa = k\lambda_2 - L$,

$$\dot{V} \leq -pv^2 + 2Lv|\tilde{\xi}| - (k\lambda_2 - L)|\tilde{\xi}|^2 + \theta(t)|\tilde{\xi}|$$

$$= -pv^2 + 2Lv|\tilde{\xi}| - \kappa|\tilde{\xi}|^2 + \theta(t)|\tilde{\xi}|$$

$$= -\begin{pmatrix} v \\ |\tilde{\xi}| \end{pmatrix}^T \begin{pmatrix} p & -L \\ -L & \kappa \end{pmatrix} \begin{pmatrix} v \\ |\tilde{\xi}| \end{pmatrix} = \rho_\kappa(v, \tilde{\xi}).$$

If $k > \bar{K}$, then $\kappa > 3L^2/p$ and hence, by lemma 2,

$\rho_\kappa(v, \tilde{\xi}) \leq -c(|v|^2 + |\tilde{\xi}|^2) = -2cV$ whenever $2V > \theta(t)^2 r(1/k)$.

* Step 4 : Bound V

Recall that $\theta(t) = |R^T \bar{f}(t, \mathbb{1}s)|$. Then,

$$\|\theta(t)\| \le \|R\| \cdot \|\bar{f}(t, \mathbb{1}s)\| = \sqrt{\textstyle\sum_i \bar{f}_i(t,s)^2} \le \sqrt{N}\, M(|s|)$$

(Assumption 1)

Also, $|\bar{f}(t,s)| = \left|\frac{1}{N}\mathbb{1}^T \bar{f}(t,s)\right| \le \frac{1}{N}\sum_i |\bar{f}_i(t,s)| \le \frac{1}{N} \cdot N \cdot M(|s|) = M(|s|)$

and hence, $|\bar{f}(t,0)| \le M(0)$

Then, by lemma 1. $\limsup_{t\to\infty} |s(t)| \le \frac{1}{p} M(0)$, i.e.

for any $\delta > 0$, $|s(t)| \le \frac{M(0)}{p} + \delta$ for sufficiently large t.

By letting $\delta \to 0$, $\bar{\theta} := \limsup_{t\to\infty} \theta(t) \le \sqrt{N}\, M\left(\frac{M(0)}{p}\right) < \infty$.

Together with such a bound of $\theta(t)$.

for any $\delta > 0$, $\exists T$ s.t. $\theta^2(t) r(1/k) \leq \bar{\theta}^2 r(1/k) + \delta$ for $t \geq T$.

That is, if $2V > Q$ for $t \geq T$, then $\dot{V} \leq -Qc$ and hence

$2V \leq Q$ be achieved in finite time, and is forward-invariant.

Thus, $\limsup_{t \to \infty} 2V(t) \leq Q$.

Since $|x_i - s| \leq |x - \mathbf{1}s| = \sqrt{2V}$,

$$\limsup_{t \to \infty} |x_i - s| \leq \bar{\theta} \sqrt{r(1/k)} \leq \sqrt{N} M\left(\frac{M(0)}{P}\right) \sqrt{r\left(\frac{1}{k\lambda_2 - L}\right)}$$

■

Question 2.

How consensus give rise to robustness of group behavior?

Q2 How consensus give rise to robustness of group behavior?

Rewrite dynamics

$$\dot{x}_i = f_i(t, x_i) + u_i = g_0(t, x_i) + \sum_{j=1}^{d} \Delta_{ij} g_j(t, x_i) + u_i$$

- g_0: nominal dynamics common to every agent.
- g_j: generating functions determine heterogenity.
- Δ_{ij}: Random variable that means weight of g_j on the agent i.
  - They are independent for all i (possibly dependent for j)

WLOG. let $E\{\Delta_{ij}\} = 0$ and $Var(\Delta_{ij}) = \sigma_j^2$

* Blended dynamics

$$\dot{s} = \bar{f}(t,s) = \frac{1}{N}\sum_{i=1}^{N} f_i(t,s) = g_0(t,s) + \sum_{j=1}^{d} \bar{\Delta}_j g_j(t,s) \quad \text{where} \quad \bar{\Delta}_j = \frac{1}{N}\sum_{i=1}^{N} \Delta_{ij}$$

Since $E\{\Delta_{ij}\} = 0$ and $\mathrm{Var}(\Delta_{ij}) = \sigma_j^2$. $E\{\bar{\Delta}_j\} = 0$ and $\mathrm{Var}(\bar{\Delta}_j) = \sigma_j^2/N$

* Expected blended dynamics

$$E\{\dot{s}\} = E\left\{g_0(t,s) + \sum_{j=1}^{d} \bar{\Delta}_j g_j(t,s)\right\} = E\{g_0(t,s)\} + \sum_{j=1}^{d} E\{\bar{\Delta}_j g_j(t,s)\}$$

$$= g_0(t,s) + \sum_{j=1}^{d} g_j(t,s) \cancel{E\{\bar{\Delta}_j\}}^{\,0} = g_0(t,s)$$

$$\dot{s}_E(t) = g_0(t, s_E), \quad s_E(0) = s(0).$$

Theorem 2) Robustness of blended dynamics

Together with the assumptions on theorem 1,

there exists a class $\mathcal{K}$-function γ_N^* s.t

$\gamma_N^* \to 0$ as $k \to \infty$

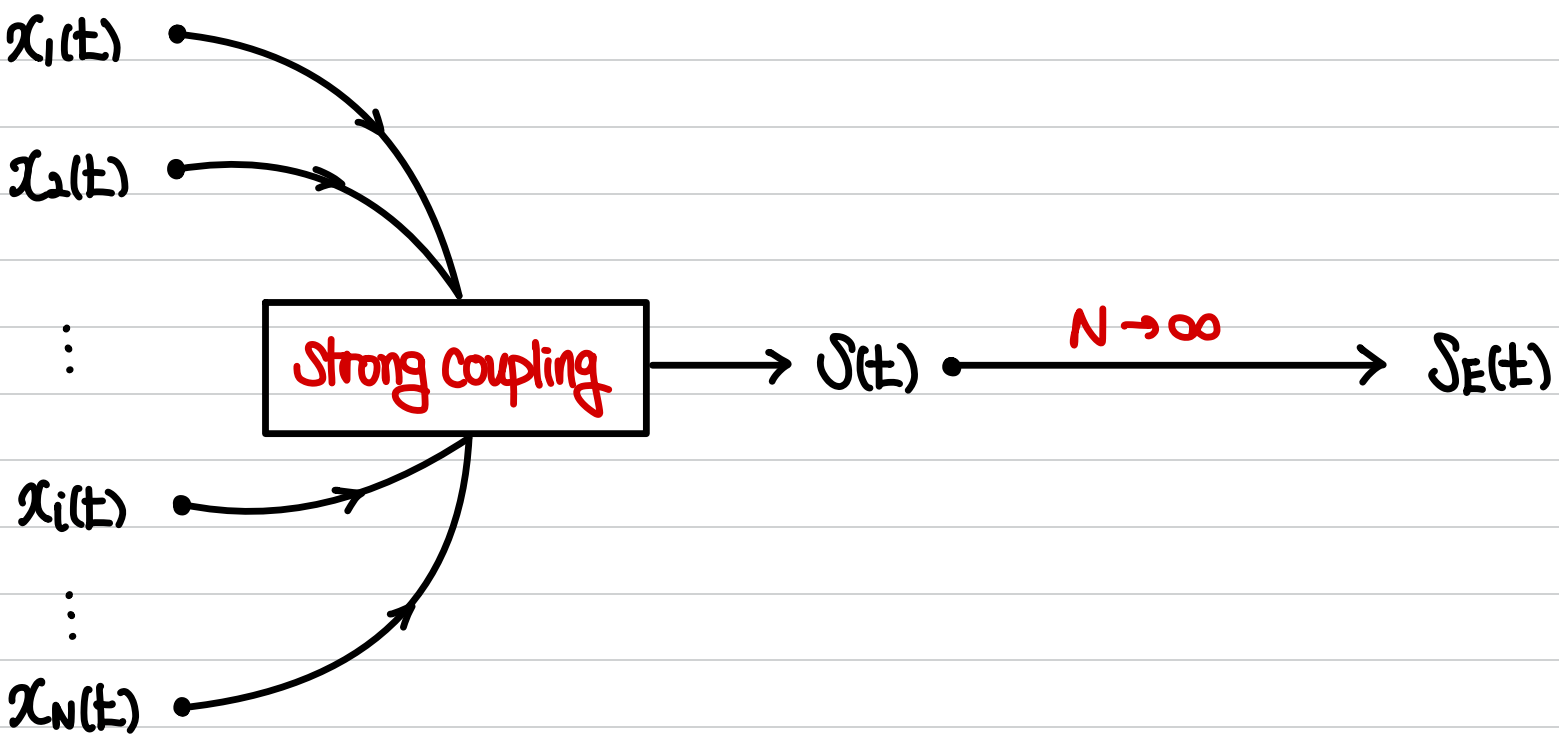
$$P\left(\limsup_{t\to\infty} |x_i(t) - s_E(t)| \leq \gamma_N^*\left(\tfrac{1}{k\lambda_2} - L\right) + \varepsilon,\ \forall k > \bar{K}_N,\ \forall i\right)$$

$$\geqslant \left(1 - \frac{d}{N^2}\right)^N + \sum_{j=1}^{d}\left(1 - \frac{4d^2\sigma_j^2 L_i^2}{p^2 N}\right)\left(1 - \frac{4d^2\sigma_j^2 M_j \left(M(0)/p\right)^2}{\varepsilon^2 p^2 N}\right) - d$$

As $N \to \infty$, RHS $\to 1$ and hence,

$$\lim_{N\to\infty} P\left(\limsup_{t\to\infty} |s(t) - s_E(t)| < \varepsilon\right) = 1$$

In short,



* Comparison table

General Dynamics structure: $\dot{x} = -kL(G)x + u_i$

	P_1	P_2	P_3
Topic	Consensus	Network controllability	Blended dynamics
Agents	identical	identical, some are leaders	Heterogeneous strong diffusive coupling
Question	System states agree?	Can we control the convergence point?	How the swarm behaves?
Answer	Yes, to average.	No, if $\exists$ symmetric network structure.	Averaged vector field.

Thank you.

It's the end of the presentation